

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment considerations.

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Layered (N-Tier)

Functional Requirements

- Submit CV
- View and Edit CV
- Answer Screening Questions
- Get Rating for CV
- Parse CV
- Compare CV with Job Requirements
- Validate and Score Applicant Responses
- Generate Percentage-Based Result

Non-Functional Requirements

- Performance

Description: The system **MUST** respond to user requests within a reasonable time frame.

- Reliability

Description: The system **MUST** be able to accurately parse and analyze CVs and provide reliable results.

- Security

Description: The system **MUST** protect user data and prevent unauthorized access.

- Privacy

Description: The system **MUST** ensure that user data is handled in accordance with applicable laws and regulations.

- Scalability

Description: The system **MUST** be able to handle a large volume of user requests and scale accordingly.

- Usability

Description: The system **MUST** provide an intuitive and user-friendly interface that is easy to navigate.

- Maintainability

Description: The system MUST be designed to be easily maintainable and updatable.

- Compatibility

Description: The system MUST be compatible with a variety of devices and browsers.

- Integration

Description: The system MUST be able to integrate with other systems and applications.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, con-

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Layered (N-Tier)";
  Property source : String = "AI";
  Component CVParser = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Parse and extract key information from the applicant's CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ScreeningQuestionValidator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Verify and grade applicant responses against an expert-generated CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CosineSimilarityCalculator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Assess the alignment between CV information and job requirements";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component JaccardSimilarityCalculator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Assess the alignment between CV information and job requirements";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RatingGenerator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Generate a percentage-based result representing the applicant's fit";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
}
```

```

}
Component CVComparisonModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Compare CV information with job requirements and generate a
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component ChatGPTValidator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Validate and score applicant responses using ChatGPT";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component SubmitCVModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allow applicants to submit their CVs for parsing and analysis";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component ViewandEditCVModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Enable applicants to view and edit their parsed CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component AnswerScreeningQuestionsModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Present applicants with screening questions defined by an ex
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component GetRatingforCVModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Provide applicants with a rating for their submitted CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment SubmitCVModule.out to CVParser.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVParser.out to CVComparisonModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVComparisonModule.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
}

```

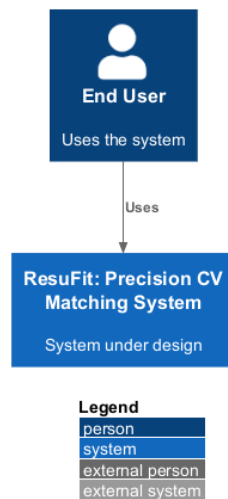
```
Attachment CVComparisonModule.out to GetRatingforCVModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment SubmitCVModule.out to ViewandEditCVModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment ViewandEditCVModule.out to SubmitCVModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment AnswerScreeningQuestionsModule.out to ScreeningQuestionValidator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment ScreeningQuestionValidator.out to RatingGenerator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment SubmitCVModule.out to ChatGPTValidator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment ChatGPTValidator.out to RatingGenerator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment SubmitCVModule.out to CosineSimilarityCalculator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment CosineSimilarityCalculator.out to CVComparisonModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment SubmitCVModule.out to JaccardSimilarityCalculator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment JaccardSimilarityCalculator.out to CVComparisonModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment AnswerScreeningQuestionsModule.out to ChatGPTValidator.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
```

```

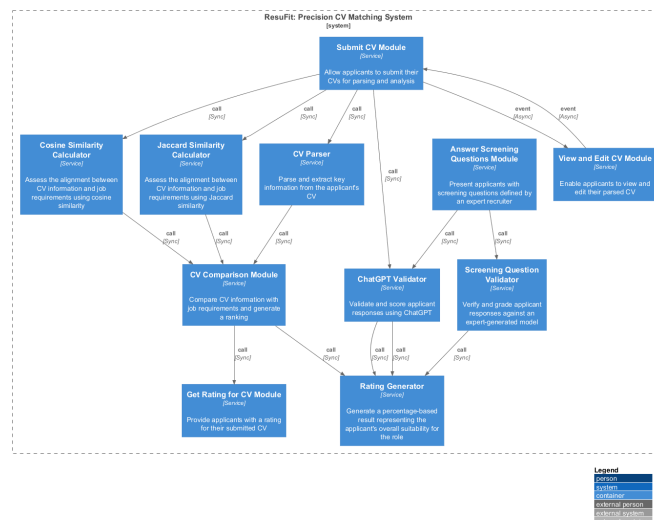
    Property backpressure : String = "bounded";
}
Attachment ChatGPTValidator.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry";
Property delivery : String = "at-most-once";
Property ordering : Boolean = true;
Property authentication : String = "centralized";
Property authorization : String = "RBAC";
}

```

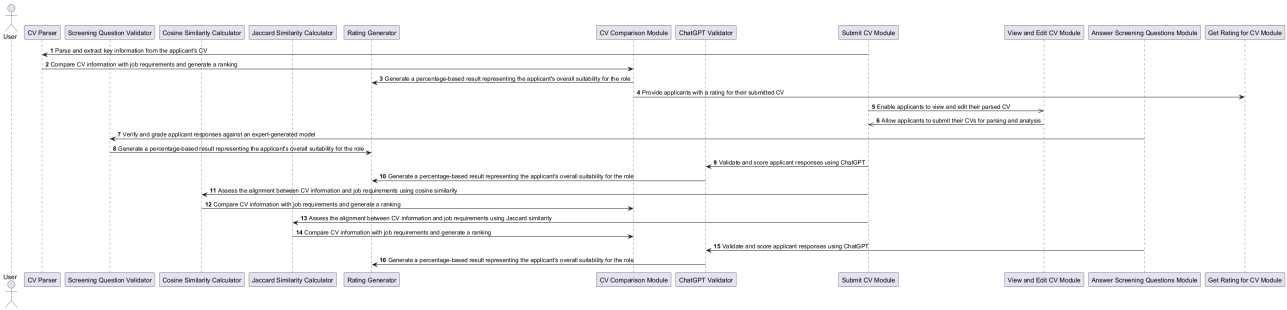
3. Context View



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

